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Artificial intelligence and stock price crash risk: Evidence from China: A pre-registered study

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ABSTRACT

This pre-registered study executes the empirical design approved in the associated pre-registered report (Bai and Zhao, 2025) to investigate the impact of artificial intelligence (AI) investment on stock price crash risk. Using China's "New-generation Artificial Intelligence Polit Zone Policy" as a quasi-natural experiment, we find robust evidence that AI investment significantly increases firms' stock price crash risk, mainly due to reduced information transparency and heightened managerial optimism. The effect is more pronounced for firms with lower levels of information transparency and tighter resource constraints. Furthermore, we also find the policy boosts firm value, suggesting that market optimism may drive short-term valuation gains at the cost of long-term stability. Overall, our findings highlight the unintended downside risks associated with AI investment, emphasizing the importance of transparency and governance in mitigating potential adverse outcomes.

1. Introduction

The rapid development of artificial intelligence (AI) has profoundly reshaped the global economy, positioning AI as a transformative general-purpose technology with far-reaching implications for firms, industries, and markets. Governments worldwide actively promote AI adoption, with China's "New-generation Artificial Intelligence Pilot Zone" policy serving as a major initiative to foster AI-driven growth. Against this backdrop, assessing the economic consequences of corporate AI investment has become both urgent and policy-relevant. Existing research has mainly focused on the real-side effects of AI, showing that adoption enhances productivity, drives innovation, and reshapes labor markets through both job displacement and creation (Cockburn et al., 2019; Acemoglu et al., 2022; Yang, 2022; Damioli et al., 2023; Babina et al., 2024). However, less is known about its capital market consequences, especially how AI investment influences investor risks and stock price stability.

Theoretically, the impact of AI investment on stock price crash risk is ambiguous. On the positive side, AI can improve information processing, enhance efficiency, and create credible growth prospects, which may reduce managers' incentives to conceal bad news and thus lower crash risk. On the negative side, the uncertainty and complexity of AI projects may amplify managerial optimism and information asymmetry, encouraging managers to withhold unfavorable information and increasing the likelihood of crashes when

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